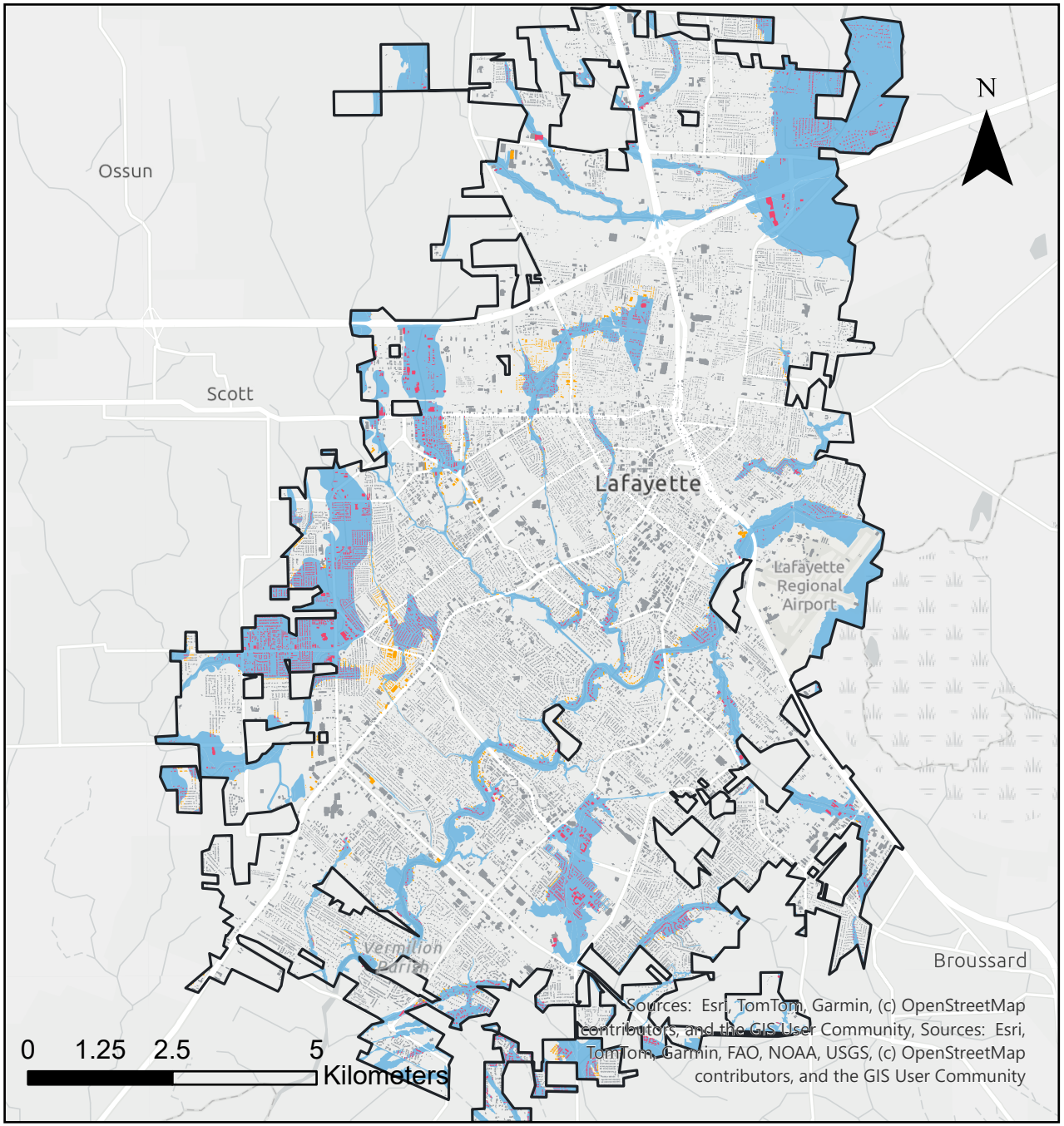
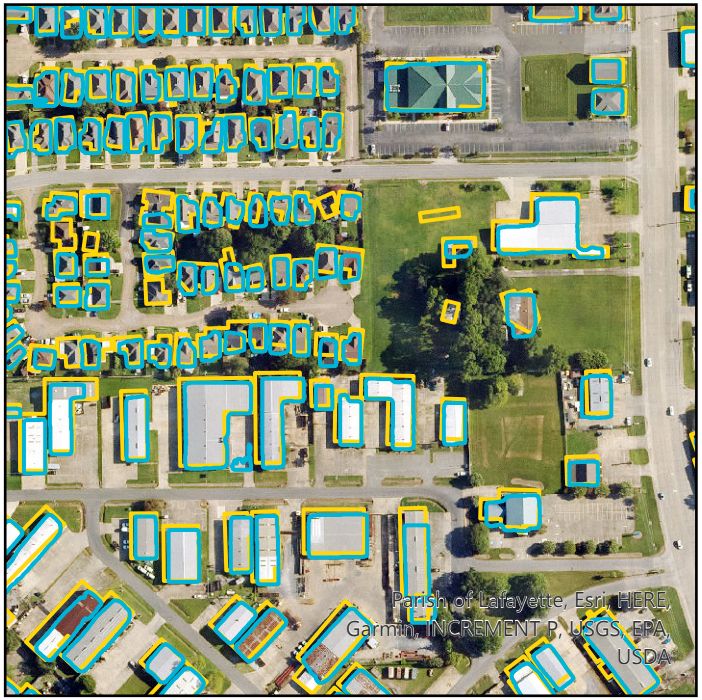
GeoAI Building Footprints & Flood Exposure

City of Lafayette, Louisiana | Esri pretrained Mask R-CNN on 2023 NAIP imagery



Detail: cyan = AI, yellow = USA Structures



City of Lafayette

Building: 0.2% flood zone

Building: 1% flood zone

Building: outside mapped flood zones

43,745 buildings detected
5,708 (13.1%) in the 1% floodplain
Precision 62%, recall 56% vs USA Structures
Blue shading: FEMA 1% annual-chance flood zone

Data: USDA NAIP 2023 (Microsoft Planetary Computer); FEMA NFHL; FEMA/ORNL USA Structures; Census 2023. NAD83 / UTM 15N. Author: Shristi Karki